

1 Introduction

This is an informal report for the acoustic signal classification project. The goal of this project is to classify acoustic signals using machine learning techniques into "Drummy" or "Tight". The dataset used for this project is divided into 2 phases: Phase 1 and Phase 2.

2 Stage 1

2.1 Data Analysis

2.1.1 Phase 1

Phase 1 consists of **3,309 recordings (2,212 drummy sounds and 1,097 tight sounds)**. Recordings in this phase have a duration of **0.75 seconds** and **36000 samples per recording**. The recordings are sampled at a rate of **48 kHz**. The dataset is not balanced, with number of drummy sounds twice that of tight sounds.

Findings

- Did a quick PCA analysis (first standardizing the data) and plotted the first 2 principal components. The 3D scatter plot (PC1 vs PC2 vs Label) shows that drummy and tight sounds are not linearly separable, and that tight sounds form a cluster in the middle of drummy sounds, while drummy sounds are more spread out.

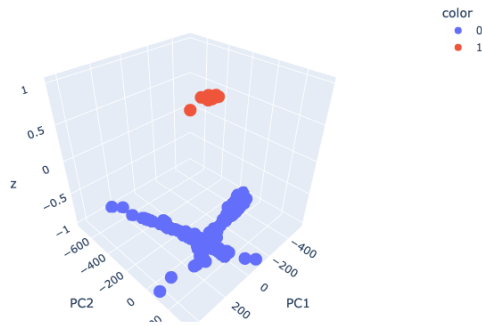
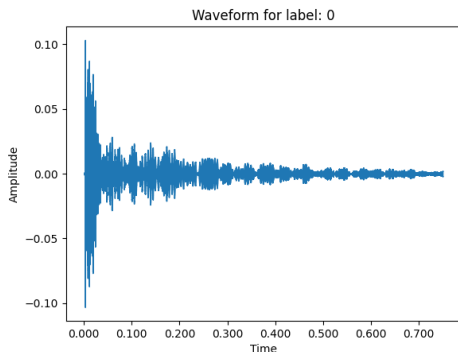
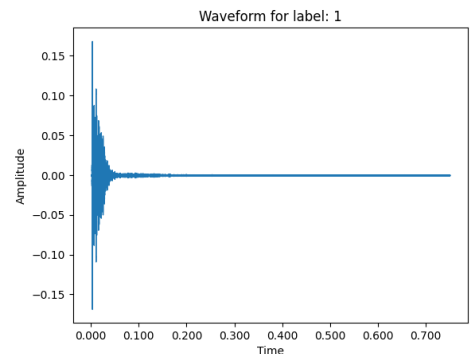


Figure 1: 2 Component PCA Analysis of Phase 1 Data

- Waveform visualization shows that drummy sounds dampen relatively slowly and have a longer decay time, while tight sounds dampen quickly and have a shorter decay time. This could suggest that sounds oscillated more in the hollow structure of the roof that produce the drummy sounds.



((a)) Drummy Waveform Visualization (Phase 1)



((b)) Tight Waveform Visualization (Phase 1)

Figure 2: Waveform visualizations for different drum styles in Phase 1

- Explained variance ratio vs Number of components gives the following results:
Number of components retaining 95% variance: 396
Number of components retaining 80% variance: 87
Number of components retaining 70% variance: 48

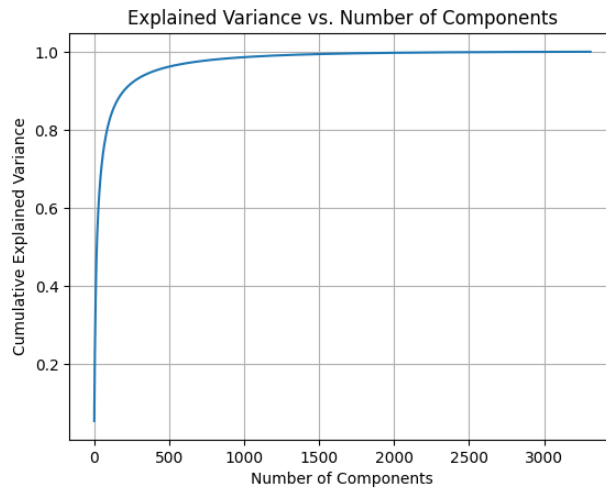


Figure 3: Explained Variance Ratio vs Number of Components (Phase 1)

2.1.2 Phase 2

Phase 2 consists of **9,923 recordings** (from which **1,483** recordings were **corrupted with NaN values**, making it 8,440 recordings; further, there are **some mis-hits** as well that need to be cleaned later for feature extraction to be done on them). Finally, there are **3,856 drummy sounds and 3,124 tight sounds (including the mis-hits)**. Each recording has 4800 samples and a duration of **0.1 seconds**. The recordings are sampled at a rate of **48 kHz**. The dataset is more balanced than Phase 1, with number of drummy sounds slightly higher than tight sounds.

Findings

- Performed a quick PCA analysis (after standardizing the data) and plotted the first two principal components. The 3D scatter plot (PC1 vs PC2 vs Label) indicates that drummy and tight sounds are not linearly separable; both classes are intermixed in the feature space.

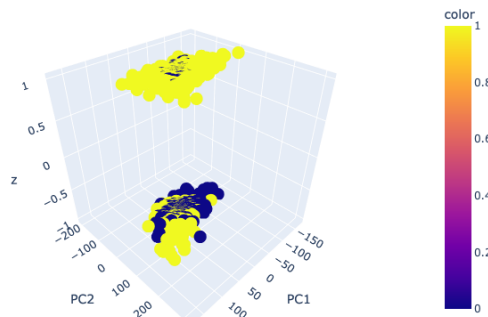
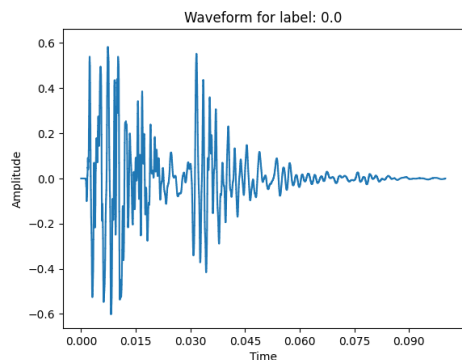
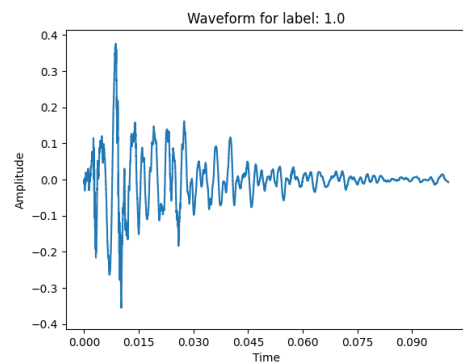


Figure 4: 2 Component PCA Analysis of Phase 2 Data

- Waveform visualizations show that drummy sounds decay more slowly and have a longer decay time, while tight sounds decay quickly and have a shorter decay time. This suggests that drummy sounds may result from greater oscillation in the hollow structure of the roof.



((a)) Drummy Waveform Visualization (Phase 2)



((b)) Tight Waveform Visualization (Phase 2)

Figure 5: Waveform visualizations for different drum styles in Phase 2

- The explained variance ratio versus number of components yields the following results:
 Number of components retaining 95% variance: 242
 Number of components retaining 80% variance: 99
 Number of components retaining 70% variance: 65

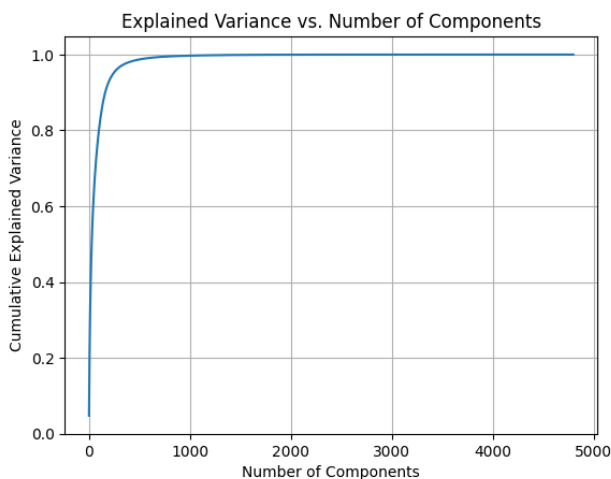


Figure 6: Explained Variance Ratio vs Number of Components (Phase 2)

3 Feature Extraction

3.1 PCA

3.1.1 Phase 1 Data

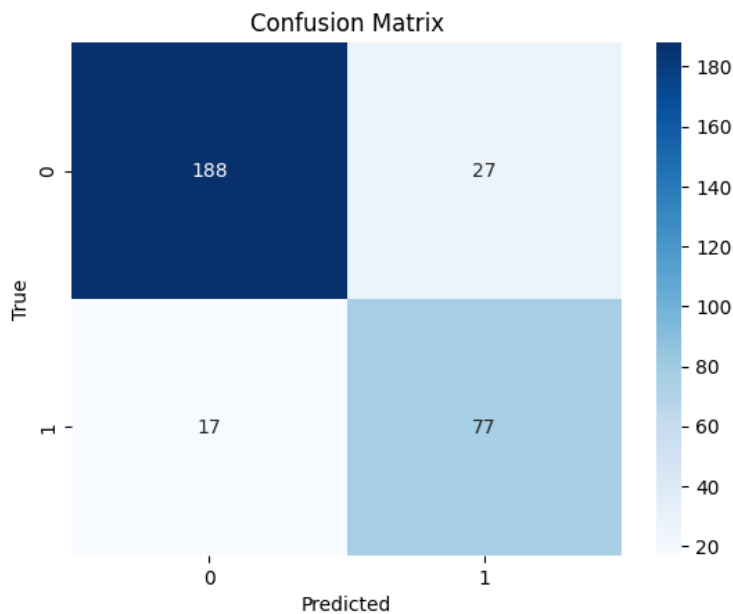
PCA / MLP Used PCA to reduce dimensionality and then classified using MLP with one hidden layer of varying number of neurons. The best results were obtained with 1000 PCA components and 64 hidden neurons with test accuracy of 0.838188. Below is the comparison chart.

Test Accuracy vs. PCA Explained Variance (facet by hidden units)



Figure 7: PCA + MLP Results (Phase 1)

PCA / SVM Used PCA to reduce dimensionality and then classified using SVM with **RBF kernel**. The best results were obtained with 1000 PCA components and a C value of 100 with test accuracy of averaging around 0.86 if we do multiple runs. Below is confusion matrix for the best run and score chart.



((a)) PCA + SVM Confusion Matrix (Phase 1)

	precision	recall	f1-score	support
0	0.92	0.87	0.90	215
1	0.74	0.82	0.78	94
accuracy			0.86	309
macro avg	0.83	0.85	0.84	309
weighted avg	0.86	0.86	0.86	309

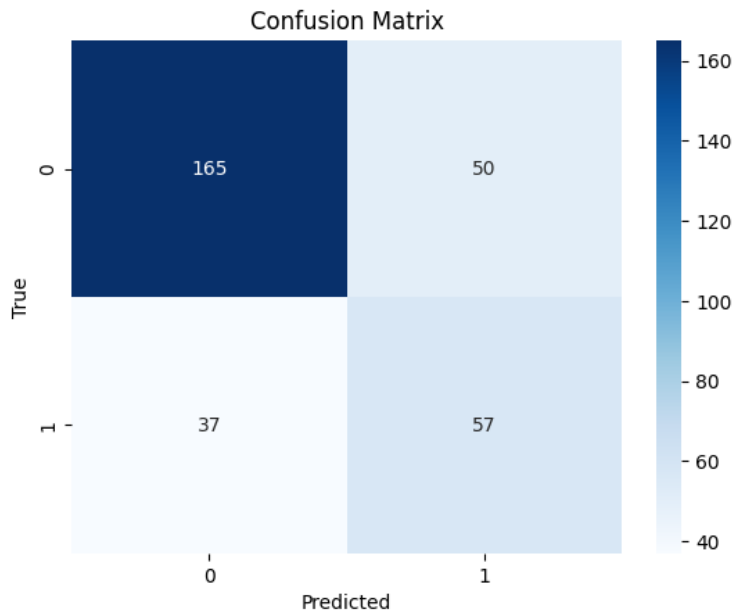
((b)) PCA + SVM Score Chart (Phase 1)

Figure 8: PCA + SVM evaluation results for Phase 1: (a) confusion matrix, (b) score chart.

3.2 LDA

3.2.1 Phase 1 Data

LDA for multiclass classification Used LDA to reduce dimensionality and classification. Below is the confusion matrix and score chart for the best run with 1 components.



((a)) LDA Confusion Matrix (Phase 1)

	precision	recall	f1-score	support
0	0.82	0.77	0.79	215
1	0.53	0.61	0.57	94
accuracy			0.72	309
macro avg	0.67	0.69	0.68	309
weighted avg	0.73	0.72	0.72	309

Accuracy: 0.7184466019417476

((b)) LDA Score Chart (Phase 1)

Figure 9: LDA evaluation results for Phase 1: (a) confusion matrix, (b) score chart.

3.3 Time-Domain Features

These features are computed on short frames of the raw waveform and capture simple temporal properties.

3.3.1 Zero Crossing Rate (ZCR)

Extracted ZCR using 128-sample frames and 32-sample hops. Classification results:

- **SVM (RBF, C=100)**: Test accuracy = 0.7799
- **MLP (1-layer)**: Test accuracy = 0.7832
- **MLP (2-layer)**: Test accuracy = 0.8058
- **MLP (3-layer)**: Test accuracy = 0.8155

3.3.2 RMS Energy

Extracted frame-wise RMS energy with 36-sample frames and 36-sample hops. Classification results:

- **SVM (RBF, C=100)**: Test accuracy = 0.8285
- **MLP (1-layer)**: Test accuracy = 0.7832
- **MLP (2-layer)**: Test accuracy = 0.8123
- **MLP (3-layer)**: Test accuracy = 0.8414

3.3.3 Amplitude Envelope

Computed the amplitude envelope on 128-sample frames with 16-sample hops. Classification results:

- **SVM (RBF, C=100)**: Test accuracy = 0.7411
- **MLP (1-layer)**: Test accuracy = 0.7638
- **MLP (2-layer)**: Test accuracy = 0.7961
- **MLP (3-layer)**: Test accuracy = 0.8123

3.4 Combined Time-Domain Features

Concatenated multiple time-domain features frame-wise and retrained the MLP.

- **Envelope + RMS**: Test accuracy = 0.8317
- **Envelope + ZCR**: Test accuracy = 0.8317
- **RMS + ZCR**: Test accuracy = 0.8447
- **Envelope + RMS + ZCR**: Test accuracy = 0.8641

3.5 Leave-One-Mine-Out Evaluation

I performed a leave-one-mine-out cross-validation using the combined feature vector (Envelope + RMS + ZCR) and an MLP with 3 layers (1024 hidden units, dropout 0.5). Figure **Leave One Out** shows test accuracy for each held-out mine.

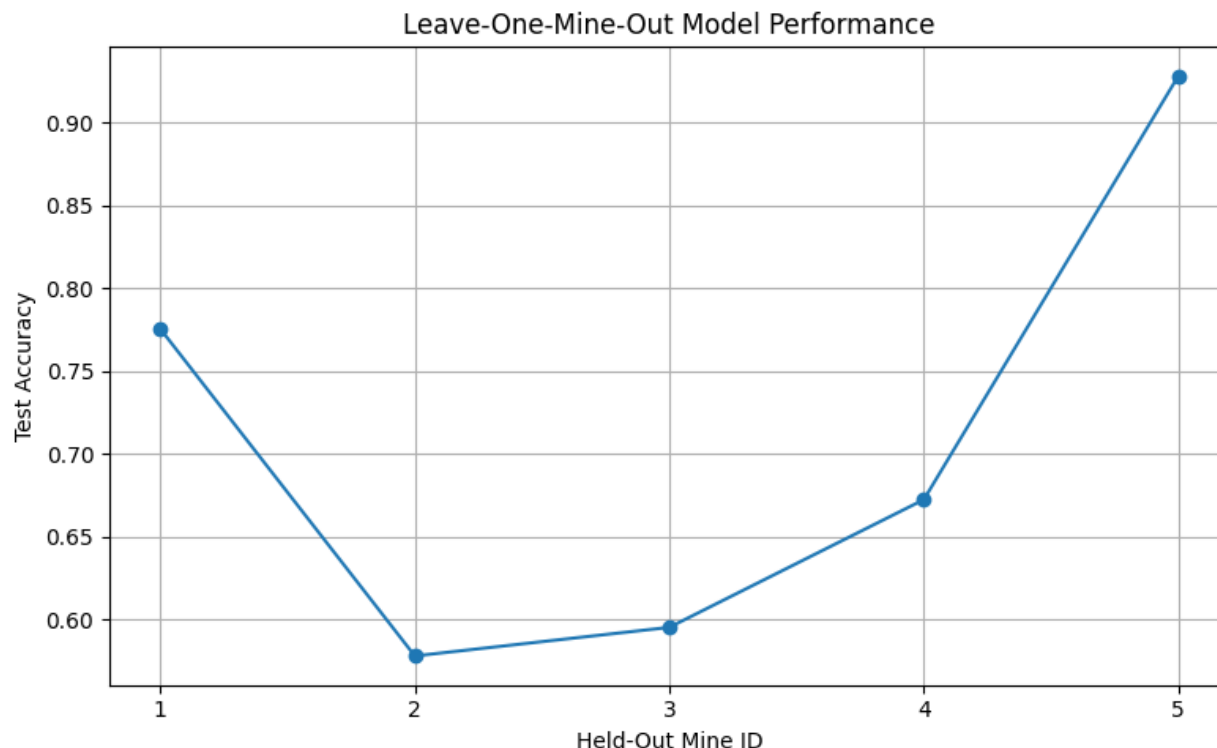


Figure 10: Leave-One-Mine-Out Performance

Average held-out accuracy: **0.7754** (drummy vs tight across mines).

3.6 Continuous Wavelet Transform (CWT)

I applied a CWT with the Complex Morlet wavelet (scales=64, 400 frames) and trained a WaveNet-style CNN.

- WaveNet classifier test accuracy = 0.6958

3.7 Discrete Wavelet Transform (DWT) Features

Computed level-5 DWT approximation coefficients using the Daubechies 4 wavelet (db4) yielding 1131 coefficients per recording. An MLP (3-layer, 256 hidden units, dropout=0.5) on these coefficients achieved:

- Test accuracy = 0.9223
- Precision/Recall/F1 for drummy (0): 0.95 / 0.94 / 0.94 (support = 215)
- Precision/Recall/F1 for tight (1): 0.86 / 0.88 / 0.87 (support = 94)

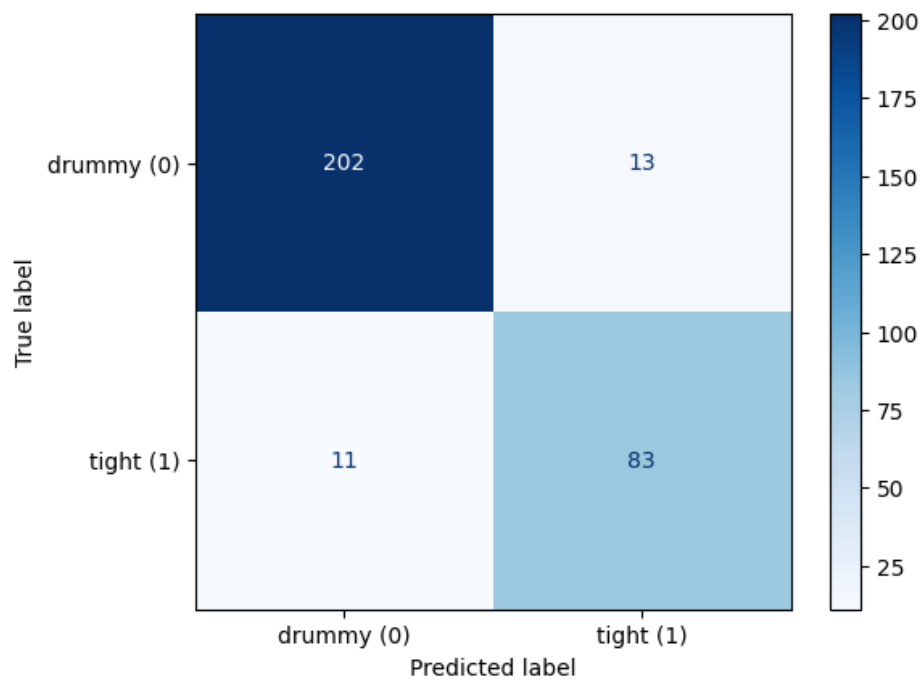


Figure 11: DWT + MLP Confusion Matrix

3.8 Combined PCA and Time-Domain Features

Concatenated PCA components (95)

- Test accuracy = 0.8835
- Precision/Recall/F1 for drummy (0): 0.91 / 0.93 / 0.92 (support = 215)
- Precision/Recall/F1 for tight (1): 0.82 / 0.79 / 0.80 (support = 94)

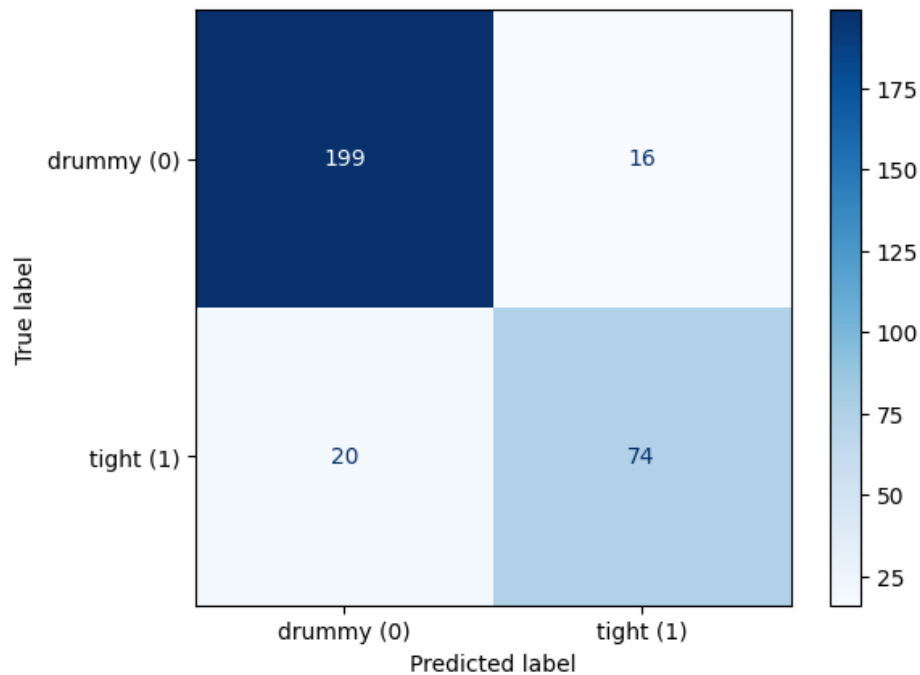


Figure 12: Combined PCA+Time Features Confusion Matrix

3.9 Raw Waveform Convolutional Classifier(Old Architecture)

Trained a residual convolutional network directly on the raw waveform (36000 samples length) with stacked ResidualBlock layers. The model achieved:

Architecture The classifier is implemented as follows:

- **ResidualBlock:** Each block applies two 1D convolutions (kernel size = 20, padding = 6), with an intervening ReLU activation. The output is optionally downsampled via a max-pooling layer (pool size and stride). A skip connection either uses identity or a 1×1 convolution (with matching stride and optional pooling) to align channel dimensions. The block output is the elementwise ReLU of the sum of main and skip paths.
- **Convolution Module:** The network begins with a Conv1d layer mapping $1 \rightarrow 2$ channels (kernel size = 10, padding = 4) and ReLU. This is followed by two ResidualBlock layers:
 - ResidualBlock($2 \rightarrow 5$, poolsize = 10, stride = 2)
 - ResidualBlock($5 \rightarrow 8$, poolsize = 10, stride = 2)

After the residual blocks, we flatten the feature map and use two fully connected layers: $704 \rightarrow 128$ with ReLU and dropout (0.5), then $128 \rightarrow 2$ output logits.

- Test accuracy = 0.9644
- Precision/Recall/F1 for drummy (0): 0.96 / 0.99 / 0.98 (support = 221)
- Precision/Recall/F1 for tight (1): 0.96 / 0.91 / 0.94 (support = 88)

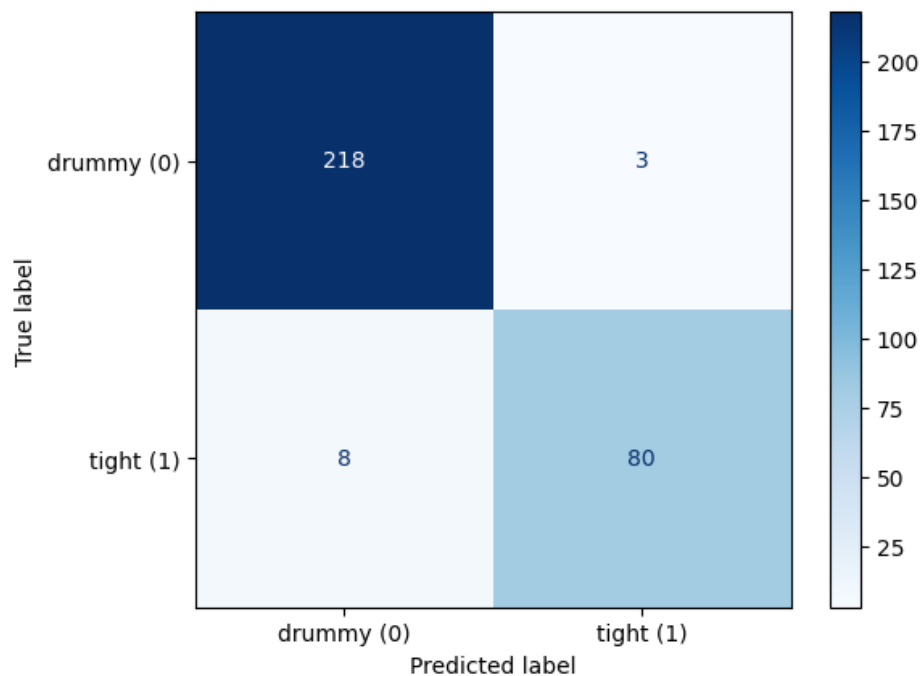


Figure 13: Raw Waveform CNN Confusion Matrix

4 Phase 1: Raw Waveform Convolutional Classifier(New Architecture)

4.1 Architecture

I implemented a deeper 1D ResNet-style CNN directly on the raw 36 000-sample waveforms:

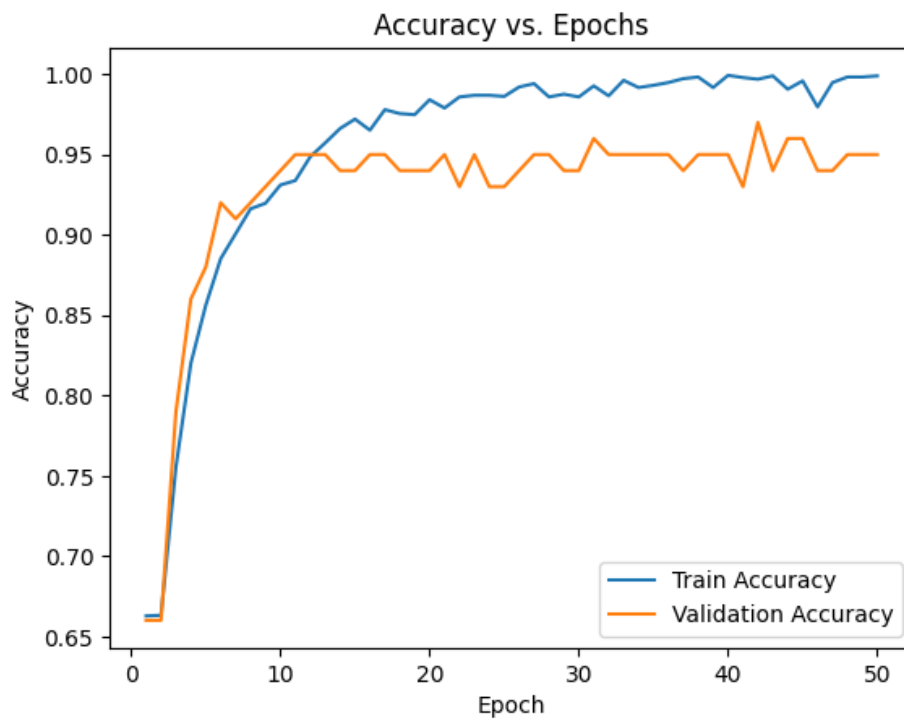


Figure 14: ResNet Test Confusion Matrix (Phase 1)

4.2 Updated Raw-Waveform Convolutional Classifier (Phase 1)

Structural Enhancements The original ResNet-style pipeline was refined with four targeted modifications, each contributing approximately +2% accuracy:

- **Kernel size expanded** from 10 to 20 samples → captures broader temporal context.
- **Tanh activations** introduced after the initial Conv1d and each ResidualBlock → smoother nonlinearity and stable gradients.
- **Dropout (0.5)** added in the fully connected head → combats overfitting on limited data.
- **Label smoothing** with $\epsilon = 0.1$ in the loss function → softens targets and improves generalization.

Block Diagram Conv1d(1→2, k=20, pad=9) → Tanh → ResidualBlock(2→5) → Tanh → ResidualBlock(5→8) → Flatten → Linear(632→64) → ear(64→2).

Performance After 20 epochs on the 3309-sample Phase 1 set:

- Test accuracy: **0.9838**
- Drummy (0): precision 1.00, recall 0.98, F1 0.99
- Tight (1): precision 0.96, recall 0.99, F1 0.97
- Conv1d(1→2) → ReLU → ResidualBlock(2→5) → ResidualBlock(5→8) → Flatten → FC(632→64) → ReLU → FC(64→2).

4.3 Results

- **Test accuracy:** 0.9709
- **Drummy (0):** precision 0.98, recall 0.98, F1 0.98 (support 221)
- **Tight (1):** precision 0.95, recall 0.94, F1 0.95 (support 88)

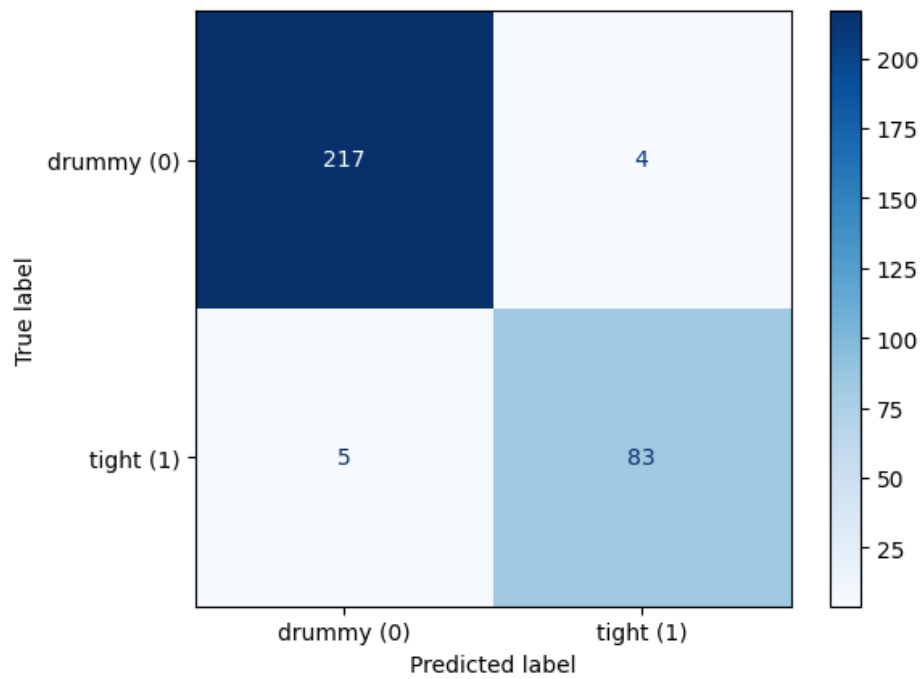


Figure 15: Simple CNN Test Confusion Matrix (Phase 1)

5 Phase 2: Raw-Waveform Classification

5.1 Convolutional Classifier (Convolution_p2)

5.1.1 Architecture

Same ResNet-style blocks adapted for 4 800-sample inputs (pool_stride=7, smaller kernels).

5.1.2 Results

- **Test accuracy:** 0.8010
- **Drummy (0):** precision 0.82, recall 0.82, F1 0.82 (support 350)
- **Tight (1):** precision 0.78, recall 0.77, F1 0.77 (support 278)

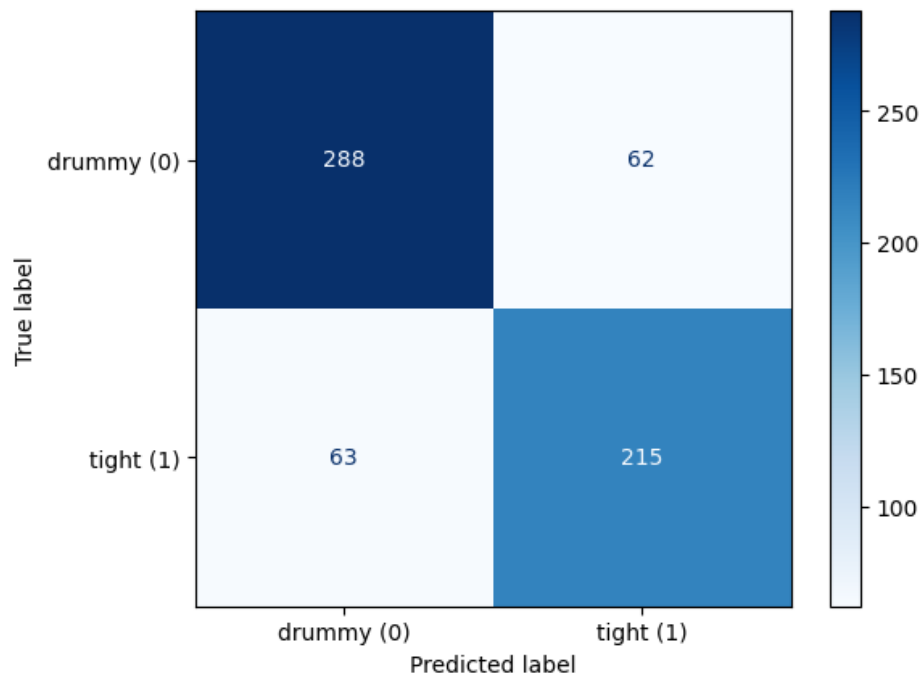


Figure 16: Raw-Waveform CNN Confusion Matrix (Phase 2)

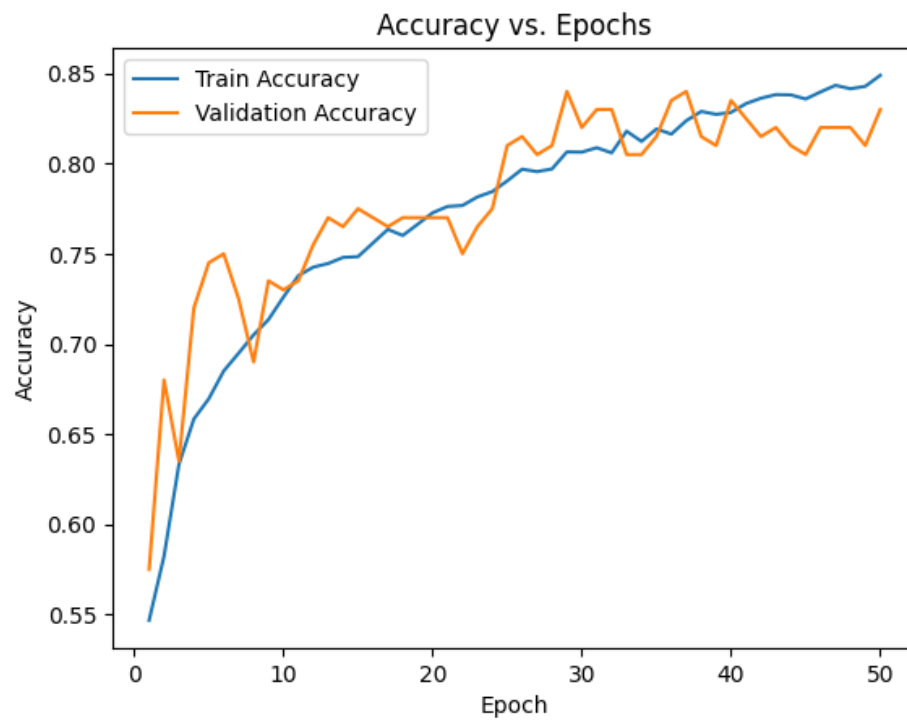


Figure 17: Raw-Waveform CNN Train vs Val Matrix (Phase 2)

5.2 Convolution + BiLSTM (ConvBiLSTM)

5.2.1 Architecture

Features are extracted via two ResidualBlock layers then fed as a sequence to a 3-layer bidirectional LSTM (hidden = 512) and finally a dense classifier.

5.2.2 Results

- **Test accuracy:** 0.8392
- **Drummy (0):** precision 0.85, recall 0.79, F1 0.82 (support 350)

- **Tight (1):** precision 0.75, recall 0.83, F1 0.79 (support 278)

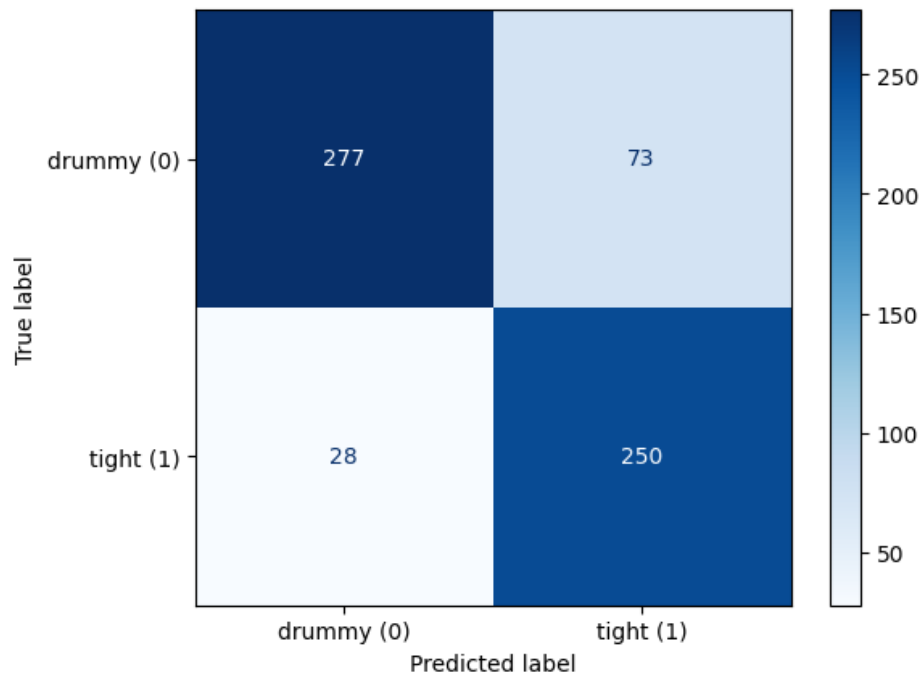


Figure 18: Conv + BiLSTM Confusion Matrix (Phase 2)

5.3 Updated Raw-Waveform Convolutional Classifier (Phase 2)

Structural Enhancements Building on the Phase 1 refinements, we further adapted channel counts and normalization:

- **Increased channels** in the first convolution: $1 \rightarrow 3 \rightarrow$ richer feature representation.
- **Tanh activations** after every convolutional stage $\rightarrow$ consistent smoothing of feature maps.
- **BatchNorm1d** applied to 5-channel output of the first ResidualBlock $\rightarrow$ stabilized feature statistics.
- **Adjusted pool_stride=2** and wavelet kernel widths $k_1 = 10, k_2 = 20$ in both ResidualBlocks $\rightarrow$ balanced downsampling.
- **Dropout (0.6)** in the classifier head $\rightarrow$ stronger regularization on the larger Phase 2 dataset.
- **Label smoothing** $\epsilon = 0.1$ carried over for consistent training behavior.

Conv1d($1 \rightarrow 3, k=20$) $\rightarrow$ Tanh $\rightarrow$ ResidualBlock($3 \rightarrow 5, \text{poolstride}=2$) $\rightarrow$ BatchNorm1d(5) $\rightarrow$ ResidualBlock($5 \rightarrow 8, \text{poolstride}=2$) $\rightarrow$ Flatten $\rightarrow$ Linear($2328 \rightarrow 512$) $\rightarrow$ ReLU $\rightarrow$ Linear($512 \rightarrow 64$) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.6) $\rightarrow$ Linear($64 \rightarrow 2$).

Performance After 20 epochs on the 8229-sample Phase 2 set:

- Test accuracy: **0.8818**
- Drummy (0): precision 0.87, recall 0.92, F1 0.90
- Tight (1): precision 0.89, recall 0.83, F1 0.86

6 Phase 2: Feature-Based Classifiers

6.1 PCA + Time-Domain Features (Envelope + ZCR + RMS)

Concatenated 95% PCA (241 dims) with envelope, ZCR and RMS (227 dims) for 468 total. Trained a 3-layer MLP (64 hidden):

- **Test accuracy:** 0.7946
- **Drummy (0):** precision 0.81, recall 0.82, F1 0.82 (support 350)
- **Tight (1):** precision 0.77, recall 0.76, F1 0.77 (support 278)

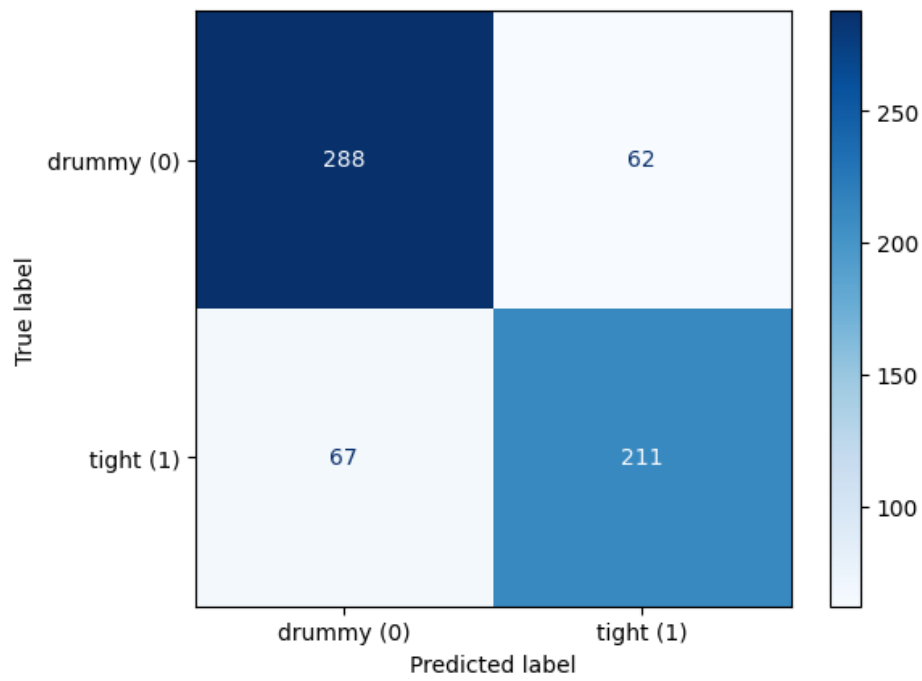


Figure 19: PCA+Time MLP Confusion Matrix (Phase 2)

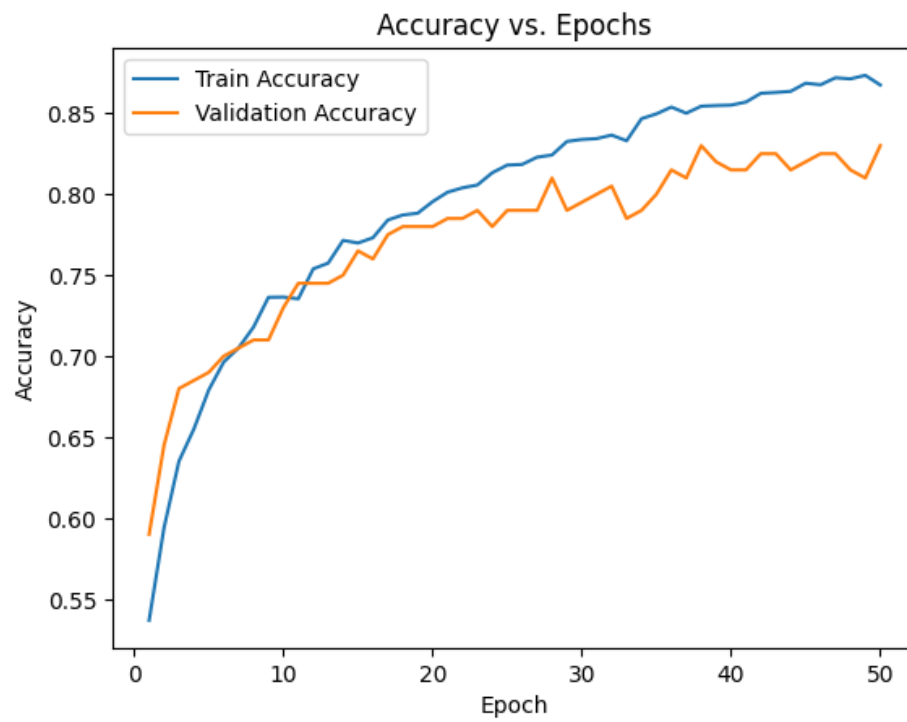


Figure 20: PCA+Time MLP Train Val (Phase 2)

6.2 MFCC-Based MLP Phase 2

Extracted 40 MFCCs ($n_{\text{fft}}=1024$, $\text{hop}=512$), trained a 3-layer MLP (64 hidden):

- **Test accuracy:** 0.9013
- **Drummy (0):** precision 0.90, recall 0.93, F1 0.91 (support 350)
- **Tight (1):** precision 0.91, recall 0.86, F1 0.89 (support 278)

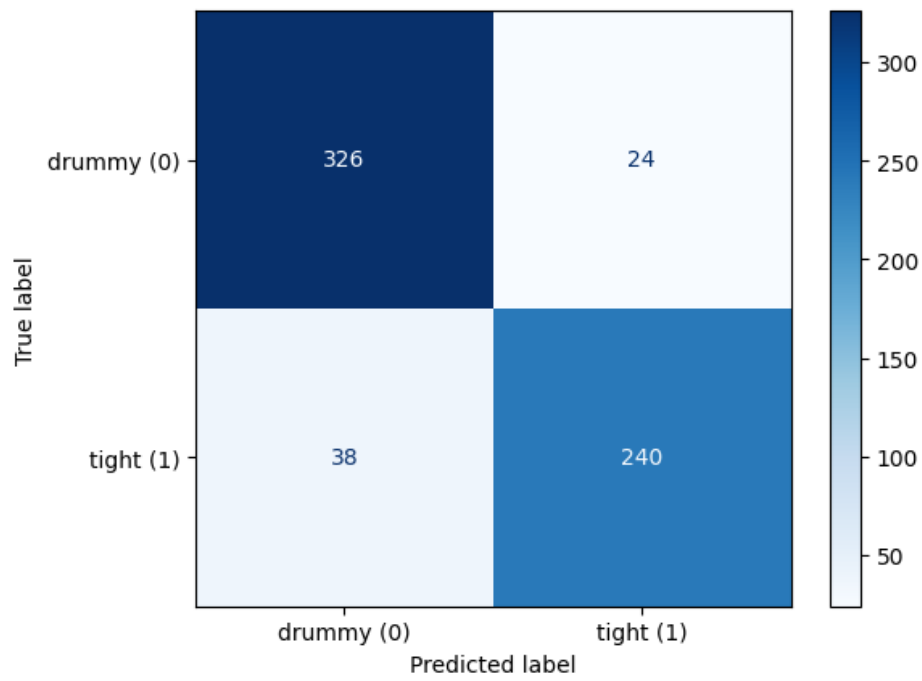


Figure 21: MFCC MLP Confusion Matrix (Phase 2)

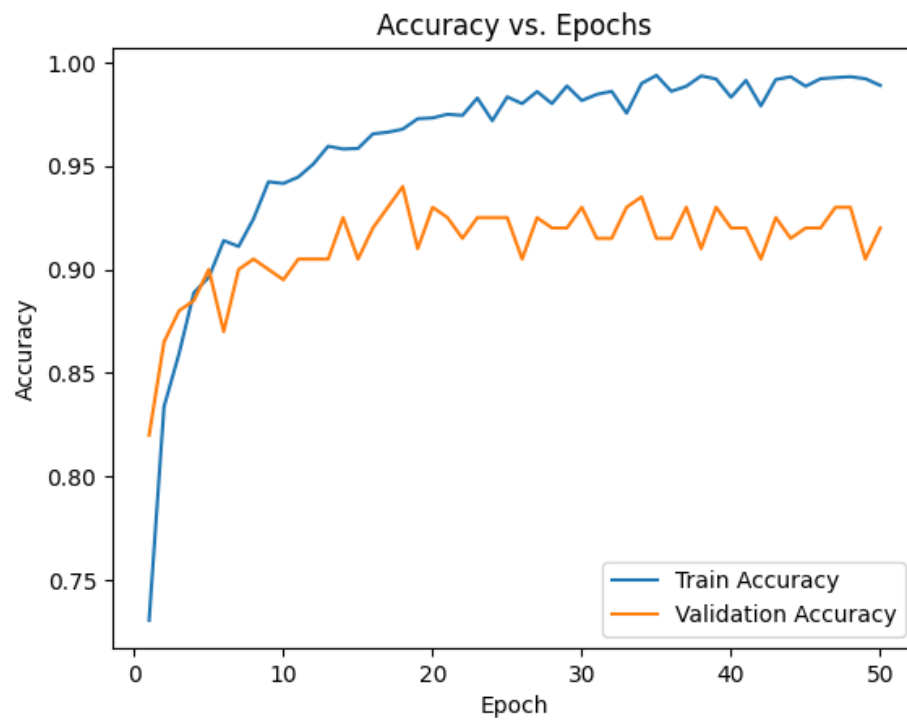


Figure 22: MFCC MLP Train Val (Phase 2)

7 Phase 2: Dual-Branch Conv + MFCC Classifier

7.1 Dual-Branch Conv + MFCC Classifier (Phase 2)

Architectural Overview Combines learned raw-waveform features with handcrafted spectral MFCCs:

Convolutional branch: $\text{Conv1d}(1 \rightarrow 2, k=20) \rightarrow \text{Tanh} \rightarrow \text{ResidualBlock}(2 \rightarrow 4, \text{poolstride}=3) \rightarrow \text{BatchNorm1d}(4) \rightarrow \text{ResidualBlock}(4 \rightarrow 6, \text{poolstride}=3)$

MFCC branch: $\text{torchaudio.transforms.MFCC}(40 \text{ coeffs}) \rightarrow \text{Flatten} ([B, 400], \text{parameters fixed})$.

Fusion and Classifier: Concat→Linear(2728→256)→ReLU→Linear(256→64)→ReLU→Dropout(0.6)→Linear(64→2).

Label smoothing $\epsilon = 0.1$ in CE Loss.

Performance After 20 epochs on Phase 2:

- Test accuracy: **0.9264**
- Drummy (0): precision 0.94, recall 0.93, F1 0.93
- Tight (1): precision 0.91, recall 0.92, F1 0.92